UNIVERSIDADE FEDERAL DE VIÇOSA
Campus de Rio Paranaíba
Sistemas de Informação
SIN 252 – Arquitetura de Computadores (T1 e T2) 2026
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Atividade 01 – Conjunto de instruções – PRÁTICA

1) Implemente em assembly do MIPS32 as seguintes funções C. Resolva primeiro em lápis e papel e, em seguida, implemente em C e simule no MARS.

a)
```
int soma_quadrados(int x, int y) {
  return (x * x) + (y * y);
}
```

b)
```
int media3(int a, int b, int c) {
  return (a + b + c) / 3;
}
```

c)
```
int dobro_sub(int a, int b) {
  return 2 * (a - b);
}
```

2) Considerando que o vetor “v” seja passado como argumento (endereço base em $a2), traduza os códigos em C a seguir. Resolva primeiro em lápis e papel e, em seguida, implemente em C e simule no MARS.

a)
```
int soma_dupla(int v[]) {
  return v[0] + v[1];
}
```

b)
```
int multiplica_espacado(int v[]) {
  return v[0] * v[4];
}
```

c)
```
int soma_n(int v[], int n) {
  int soma = 0;
  for (int i = 0; i < n; i++)
    soma += v[i];
  return soma;
}
```

3) Traduza os códigos em C contendo estruturas de decisão e lações de repetição para assembly do MIPS. Resolva primeiro em lápis e papel e, em seguida, implemente em C e simule no MARS.

a)
```
int abs(int x) {
  if (x < 0)
    return -x;
  return x;
}
```

b)
```
int max(int a, int b) {
  if (a > b)
    return a;
  return b;
}
```

c)
```
int diferenca_pos(int a, int b) {
  if (a > b)
    return a - b;
  else
    return b - a;
}
```

d)
```
int valida_primeiro(int v[]) {
  if (v[0] > 10)
    return 1;
  return 0;
}
```

e)
```
int conta_pares(int v[], int n) {
    int cont = 0;
    for (int i = 0; i < n; i++)
        if (v[i] % 2 == 0)
            cont++;
    return cont;
}
```

f)
```
int multiplica_ate_n(int v[], int n) {
  int res = 1, i = 0;
  while (i < n) {
    res *= v[i];
    i++;
  }
  return res;
}
```

INSTRUÇÕES:

A entrega deve ser realizada em um arquivo compactado no formato ZIP, nomeado obrigatoriamente da seguinte forma:

`sin252_T<turma>_<matrícula>.zip`

Exemplo: O aluno com matrícula 9991, matriculado na Turma 2, deverá entregar o arquivo:

`sin252_T2_9991.zip`

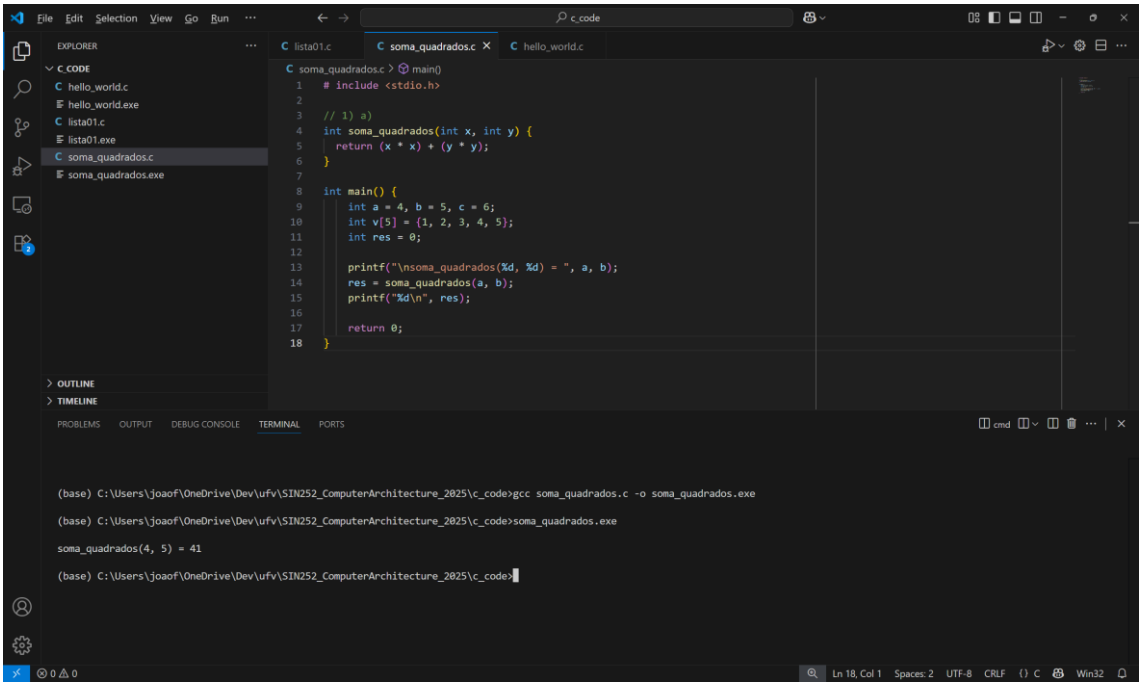
Data da entrega: 27/09/2026, via PVANet Moodle.

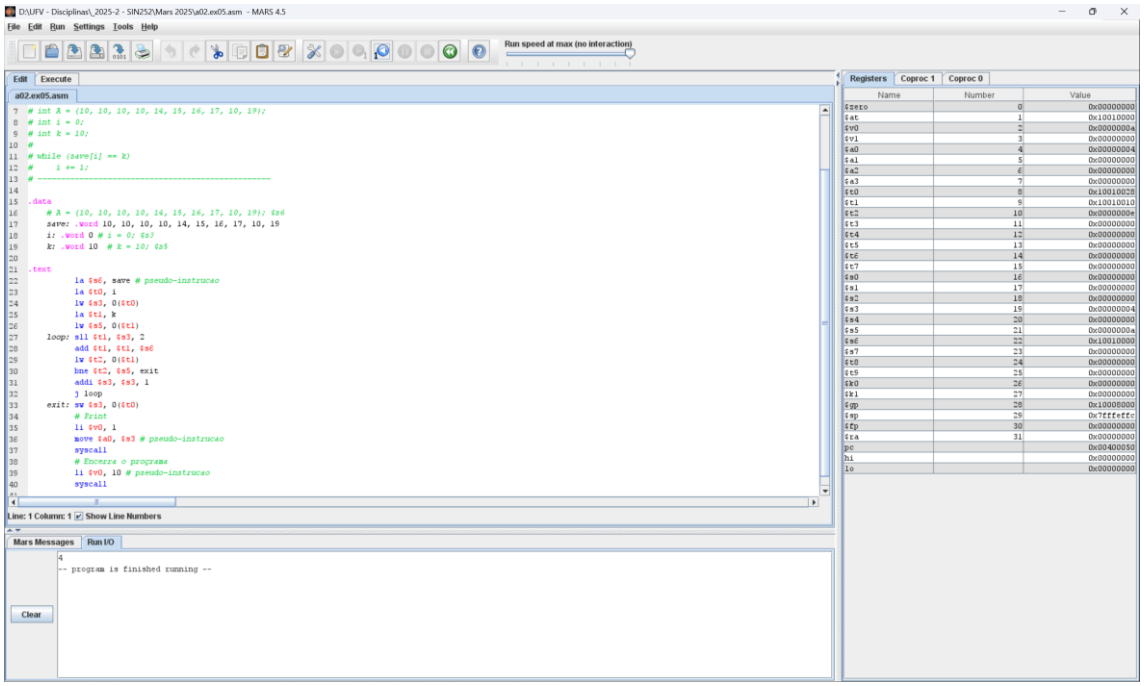
Atenção: Entregar na tarefa correta da sua turma, T1 ou T2.

O arquivo ZIP deverá conter:

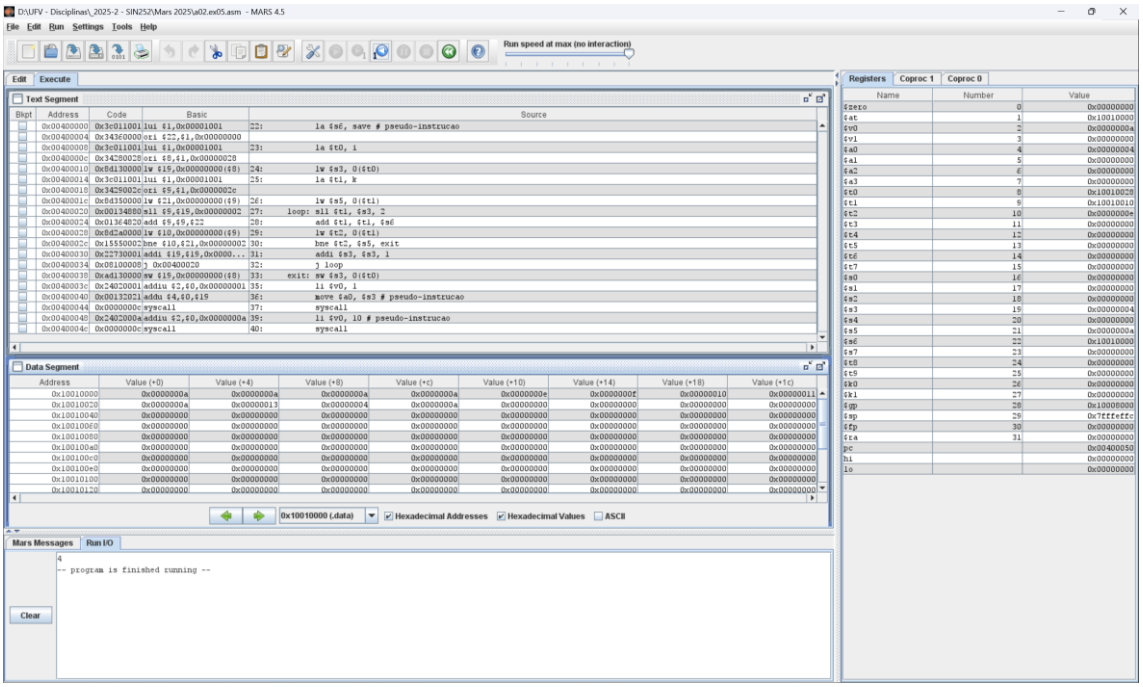
1. Arquivos fonte em C: um arquivo .c correspondente a cada item da lista.
2. Arquivos em Assembly: um arquivo .asm correspondente a cada item da lista.
3. Arquivo PDF: um único arquivo .pdf contendo:
    - Um print (captura de tela) da execução de cada programa em C, mostrando o código-fonte e o terminal no mesmo print.
    - Dois prints (captura de tela) da execução de cada programa em Assembly (tela do MARS).
        1. O código (janela Edit).
        2. Memória e programa em montado (janela Execute).

Exemplo dos prints:


File Edit Selection View Go Run
c_code
EXPLORER
C_CODE
hello_world.c
hello_world.exe
lista01.c
lista01.exe
soma_quadrados.c
soma_quadrados.exe
lista01.c
soma_quadrados.c
hello_world.c
soma_quadrados.c > main()
include <stdio.h>
// 1) a)
int soma_quadrados(int x, int y) {
return (x * x) + (y * y);
}
int main() {
int a = 4, b = 5, c = 6;
int v[5] = {1, 2, 3, 4, 5};
int res = 0;
printf("\nsoma_quadrados(%d, %d) = ", a, b);
res = soma_quadrados(a, b);
printf("%d\n", res);
return 0;
}
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
cmd
(base) C:\Users\joaof\OneDrive\Dev\ufv\SIN252_ComputerArchitecture_2025\c_code>gcc soma_quadrados.c -o soma_quadrados.exe
(base) C:\Users\joaof\OneDrive\Dev\ufv\SIN252_ComputerArchitecture_2025\c_code>soma_quadrados.exe
soma_quadrados(4, 5) = 41
(base) C:\Users\joaof\OneDrive\Dev\ufv\SIN252_ComputerArchitecture_2025\c_code>
0 0
Ln 18, Col 1 Spaces: 2 UTF-8 CRLF {} C Win32


D:\UFV - Disciplinas_2025-2 - SIN252\Mars 2025\a02.ex05.asm - MARS 4.5
File Edit Run Settings Tools Help
Run speed at max (no interaction)
Edit Execute
a02.ex05.asm
int A = {10, 10, 10, 10, 14, 15, 16, 17, 10, 19};
int i = 0;
int k = 10;
#
while (save[i] == k)
i += 1;

.data
A = {10, 10, 10, 10, 14, 15, 16, 17, 10, 19}: $s6
save: .word 10, 10, 10, 10, 14, 15, 16, 17, 10, 19
i: .word 0 # i = 0; $s3
k: .word 10 # k = 10; $s5
.text
la $s6, save # pseudo-instrucao
la $t0, i
lw $s3, 0($t0)
la $t1, k
lw $s5, 0($t1)
loop: sll $t1, $s3, 2
add $t1, $t1, $s6
lw $t2, 0($t1)
bne $t2, $s5, exit
addi $s3, $s3, 1
j loop
exit: sw $s3, 0($t0)
Print
li $v0, 1
move $a0, $s3 # pseudo-instrucao
syscall
Encerra o programa
li $v0, 10 # pseudo-instrucao
syscall
Line: 1 Column: 1 Show Line Numbers
Mars Messages Run I/O
4
-- program is finished running --
Clear
Registers Coproc 1 Coproc 0
Name Number Value
$zero 0 0x00000000
$at 1 0x10010000
$v0 2 0x0000000a
$v1 3 0x00000000
$a0 4 0x00000004
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x10010028
$t1 9 0x10010010
$t2 10 0x0000000e
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000000
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000004
$s4 20 0x00000000
$s5 21 0x0000000a
$s6 22 0x10010000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400050
hi 0x00000000
lo 0x00000000


D:\UFV - Disciplinas_2025-2 - SIN252\Mars 2025\a02.ex05.asm - MARS 4.5
File Edit Run Settings Tools Help
Run speed at max (no interaction)
Edit Execute
Text Segment
Bkpt Address Code Basic Source
0x00400000 0x3c011001 lui $1,0x00001001 22: la $s6, save # pseudo-instrucao
0x00400004 0x34360000 ori $22,$1,0x00000000
0x00400008 0x3c011001 lui $1,0x00001001 23: la $t0, i
0x0040000c 0x34280028 ori $8,$1,0x00000028
0x00400010 0x8d130000 lw $19,0x00000000($8) 24: lw $s3, 0($t0)
0x00400014 0x3c011001 lui $1,0x00001001 25: la $t1, k
0x00400018 0x34290032 ori $9,$1,0x00000032
0x0040001c 0x8d350000 lw $21,0x00000000($9) 26: lw $s5, 0($t1)
0x00400020 0x00134880 sll $9,$19,0x00000002 27: loop: sll $t1, $s3, 2
0x00400024 0x01364820 add $9,$9,$22 28: add $t1, $t1, $s6
0x00400028 0x8d2a0000 lw $10,0x00000000($9) 29: lw $t2, 0($t1)
0x0040002c 0x15550002 bne $10,$21,0x00000002 30: bne $t2, $s5, exit
0x00400030 0x22730001 addi $19,$19,0x0000... 31: addi $s3, $s3, 1
0x00400034 0x08100008 j 0x00400020 32: j loop
0x00400038 0xad130000 sw $19,0x00000000($8) 33: exit: sw $s3, 0($t0)
0x0040003c 0x24020001 addiu $2,$0,0x00000001 35: li $v0, 1
0x00400040 0x00132021 addu $4,$0,$19 36: move $a0, $s3 # pseudo-instrucao
0x00400044 0x0000000c syscall 37: syscall
0x00400048 0x2402000a addiu $2,$0,0x0000000a 39: li $v0, 10 # pseudo-instrucao
0x0040004c 0x0000000c syscall 40: syscall
Data Segment
Address Value (+0) Value (+4) Value (+8) Value (+c) Value (+10) Value (+14) Value (+18) Value (+1c)
0x10010000 0x0000000a 0x0000000a 0x0000000a 0x0000000a 0x0000000e 0x0000000f 0x00000010 0x00000011
0x10010020 0x0000000a 0x00000013 0x00000004 0x0000000a 0x00000000 0x00000000 0x00000000 0x00000000
0x10010040 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010060 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010080 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x100100a0 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x100100c0 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x100100e0 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010100 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010120 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000
0x10010000 (.data) Hexadecimal Addresses Hexadecimal Values ASCII
Mars Messages Run I/O
4
-- program is finished running --
Clear
Registers Coproc 1 Coproc 0
Name Number Value
$zero 0 0x00000000
$at 1 0x10010000
$v0 2 0x0000000a
$v1 3 0x00000000
$a0 4 0x00000004
$a1 5 0x00000000
$a2 6 0x00000000
$a3 7 0x00000000
$t0 8 0x10010028
$t1 9 0x10010010
$t2 10 0x0000000e
$t3 11 0x00000000
$t4 12 0x00000000
$t5 13 0x00000000
$t6 14 0x00000000
$t7 15 0x00000000
$s0 16 0x00000000
$s1 17 0x00000000
$s2 18 0x00000000
$s3 19 0x00000004
$s4 20 0x00000000
$s5 21 0x0000000a
$s6 22 0x10010000
$s7 23 0x00000000
$t8 24 0x00000000
$t9 25 0x00000000
$k0 26 0x00000000
$k1 27 0x00000000
$gp 28 0x10008000
$sp 29 0x7fffeffc
$fp 30 0x00000000
$ra 31 0x00000000
pc 0x00400050
hi 0x00000000
lo 0x00000000